

Current thru a diode

$$I = I_0 \left[e^{\frac{eV}{\alpha k_B T}} - 1 \right]$$

★ Move to other section.

$\alpha \rightarrow$ diode ideality factor

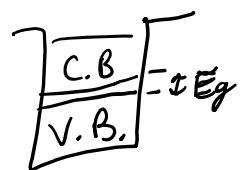
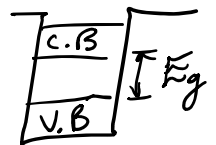
① $\alpha < 2$
 $\nwarrow$
perfect diode

Single crystal Junction (Si) - homojunction

multi crystal Junction (GaAs) - heterojunction

band gap energy

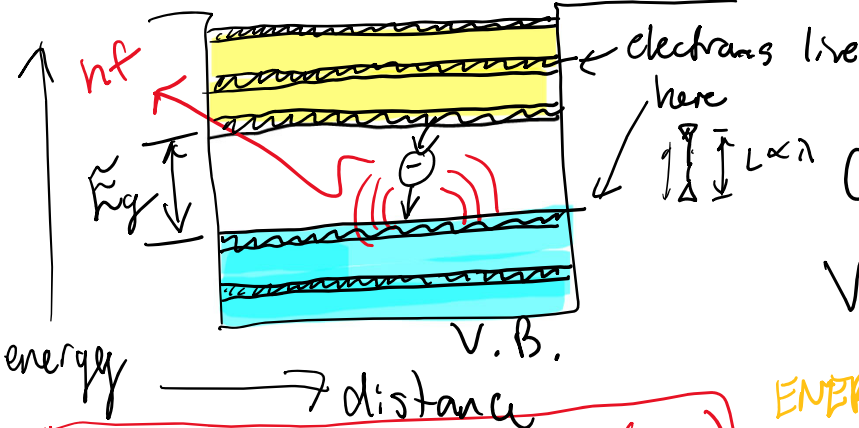
$$E_g(\text{S.C.}) > \begin{cases} \text{or} \\ \gg \end{cases} E_g(\text{m.c.})$$



* Band gap energy is the energy required for an electron to move across the gap

C.B.

ENERGY IN



C.B. = conduction Band

V.B. = Valence Band

① Vibration Energy (Loss)
 $\nwarrow$
 Light Energy - Radiation

ENERGY IN

1) Electricity ✓

2) Heat ✓

3) Light ✓

4) Force

Total interaction in a quantum well

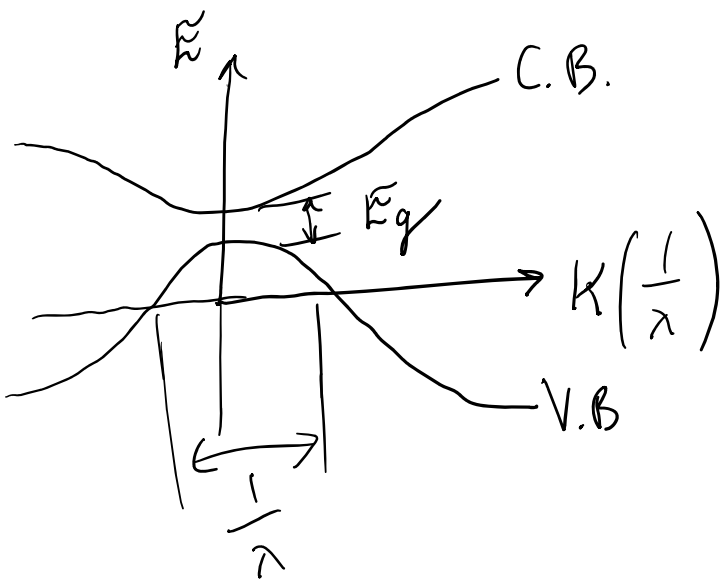
$$= \text{Non-radiating} + \text{Radiating}$$

$$= \gamma + \gamma_{\text{light}}$$

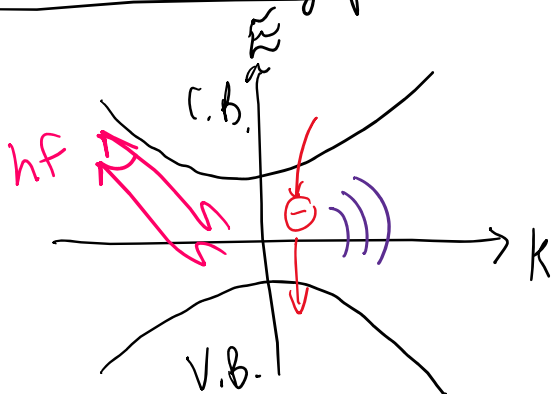
in optics "Non-radiating"
means vibrational heat

NCI near-field

Momentum Plots $\rightarrow$ Momentum $(P) = \hbar(k)$
 $\sim$ is the wave number $(\frac{1}{\lambda})$



Direct Bandgap (red LEDs)



(Most Laser diodes)

Indirect Band Gap (Blue)

